

Next-Day Mean Temperature Forecasting with the Weather Prediction Dataset

Task Report

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This report describes the problem formulation, data, model design, evaluation protocol, experimental results.

The corresponding code and outputs are included in the submitted archive.

Abstract

This report formulates and evaluates a next-day weather prediction task using the Weather Prediction Dataset. The dataset is provided via the online link: [florian-huber/weather_prediction_dataset](https://florian-huber.github.io/weather_prediction_dataset/): Create a dataset for teaching machine learning and deep learning

The objective is to predict the daily mean temperature at Basel on day $t+1$ from observations available through day t at 18 European stations. The task is framed as supervised regression with a one-day forecast horizon. The input representation combines current-day observations from all stations, calendar seasonality, lagged and rolling temperature features for Basel, and lagged mean temperature from the other stations. Four model families are compared with two baselines under a chronological evaluation protocol that never uses future information: ridge regression, random forest, histogram-based gradient boosting, and a multi-layer perceptron. Hyper-parameters are tuned on the 2007-2008 validation period, the final model family is selected on validation performance, and the selected model is refitted on training plus validation data before a single evaluation on the 2009 test period.

Gradient boosting achieved a mean absolute error of 1.228 C and a root mean squared error of 1.619 C, compared with 1.667 C and 2.188 C for a persistence baseline. This is a 26.3 percent reduction in mean absolute error. An expanding-window backtest that retrains before each month of 2009 reproduces the same conclusion, which further implied our results. Besides, a physical-range audit found 36 impossible source values, even if the dataset is claimed to be cleaned already. Hence cleaning them removed a severe validation outlier in the neural network and left the selected gradient-boosting model unchanged. The report documents the problem formulation, data, model, evaluation protocol, results, limitations, and reproducibility steps.

1. Problem Formulation

Weather prediction covers many targets, spatial scales, and lead times. This project defines one precise, verifiable forecasting problem: predict the daily mean temperature at Basel, Switzerland, one day ahead, using weather observations that are already available at the time of prediction.

1.1 Prediction setup

Let t index a calendar day. The feature vector x_t is constructed from all observations available through day t , including current-day values, lagged values, and rolling summaries. The target is $y_{(t+1)}$, the mean temperature at Basel on day $t+1$. The model learns a function f such that $y_{(t+1)}$ is approximated by $f(x_t)$. No observation from day $t+1$ or later is used as an input. This is a one-day-ahead forecast under the assumption that all station observations for day t are available when the prediction is made.

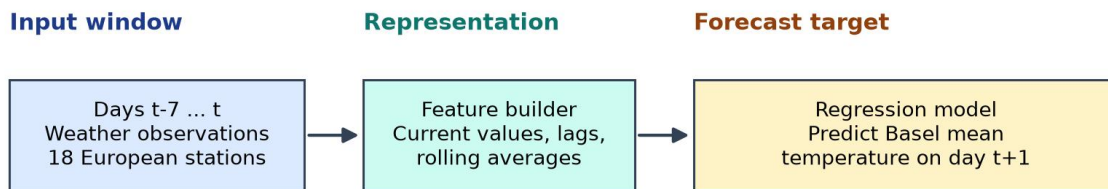
Basel was selected as the target station because all 18 stations provide temperature records, which makes the target well defined and comparable across the whole dataset. Mean temperature is continuous, interpretable, and has strong and physically meaningful baselines. The same framework could be applied to minimum or maximum temperature, precipitation, or another station without changing the evaluation protocol.

1.2 Formal objective

The learning problem is supervised regression. Given a training sample of paired observations (x_t , y_{t+1}), the objective is to minimise the expected absolute error on unseen future days. Mean absolute error (MAE) is used as the primary selection criterion because it is expressed in degrees Celsius and is robust to large individual errors. Root mean squared error (RMSE) is also reported because it penalises large errors more strongly. The coefficient of determination, R^2 , is reported for comparison with common regression practice.

1.3 Evaluation principles

Three principles guide the evaluation. First, all splits are chronological: the model is trained on the past, tuned on a later period, and tested on the most recent period. Second, the split is defined by the forecast target date, so a target from the validation or test period cannot enter the fitting data through a one-day boundary. Third, model selection uses validation performance only; the test period is held out until the final evaluation. A persistence baseline and a monthly climatology baseline are included so that the value added by machine learning is explicit.



One-day-ahead forecast: use data available through day t , predict day $t+1$

Figure 1. Problem formulation. Information flows from observations available through day t to a one-day-ahead prediction for day $t+1$.

2. Dataset

2.1 Source and scope

The project uses the Weather Prediction Dataset (Huber et al., 2022a), introduced as a teaching resource by Huber et al. (2022b) and available at https://github.com/florian-huber/weather_prediction_dataset. The underlying station observations come from the European Climate Assessment & Dataset (Klein Tank et al., 2002).

This dataset is designed for machine learning and deep learning purposes. It contains daily observations from 18 European locations between 1 January 2000 and 1 January 2010. The released table has 3,654 daily rows and 163 weather feature columns. All 18 stations provide mean, minimum, and maximum temperature. Additional variables such as cloud cover, humidity, pressure, global radiation, precipitation, sunshine, wind speed, and wind gust are available wherever the source records allowed it.

Table 1. Dataset summary used in this project.

Property	Value
Source	Weather Prediction Dataset (GitHub)
Number of daily records	3,654
Number of weather feature columns	163
Number of stations	18
Start date	2000-01-01
End date	2010-01-01
Missing values in released CSV	0
Sentinel values (-9999) in released CSV	0
Physical-range violations found	36
Duplicate dates	0

2.2 Variables and units

All selected locations provide temperature. Other variables are present for a subset of stations. The dataset authors already converted values to intuitively readable units, which keeps the features interpretable. Table 2 summarises the variable groups and units used in the report.

Table 2. Variable groups and physical units.

Variable group	Column suffix	Unit
Mean temperature	_temp_mean	degrees Celsius
Maximum temperature	_temp_max	degrees Celsius
Minimum temperature	_temp_min	degrees Celsius
Cloud cover	_cloud_cover	oktas
Global radiation	_global_radiation	100 W/m ²
Humidity	_humidity	percent
Sea-level pressure	_pressure	1000 hPa
Precipitation	_precipitation	centimetres
Sunshine	_sunshine	hours
Wind speed	_wind_speed	m/s
Wind gust	_wind_gust	m/s

2.3 Data quality and limitations

The released CSV contains no NaN entries and no -9999 sentinel values. A physical-range audit nevertheless found 36 values that are impossible or sentinel-like: four pressure values below the plausible range, including TOURS_pressure = 0.0003 on 2008-12-30 and STOCKHOLM_pressure = -0.099 on three dates; three cloud-cover values outside 0-8 oktas, including -99; and 29 negative sunshine values. The pipeline marks these values as invalid and replaces them with causal forward-filled values, using only past observations. The full list is saved to outputs/metrics/data_quality_report.csv. The dataset authors also removed columns with more than 5 percent invalid entries and replaced remaining invalid entries with mean values before release, so missingness is partly masked rather than modelled. The data are daily station observations rather than gridded reanalysis fields, and the record covers roughly ten years, which limits the amount of independent seasonal variation available for learning. These points are revisited in the limitations section.

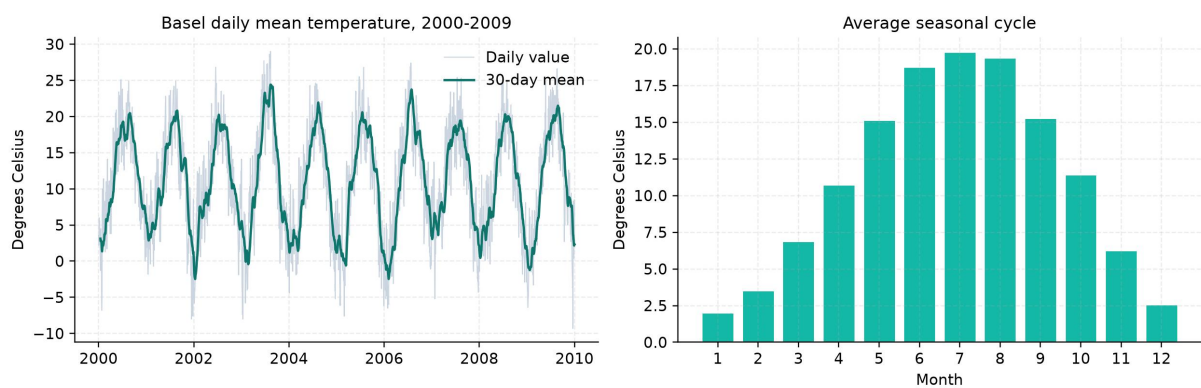


Figure 2. Basel daily mean temperature and its seasonal cycle across the available record.

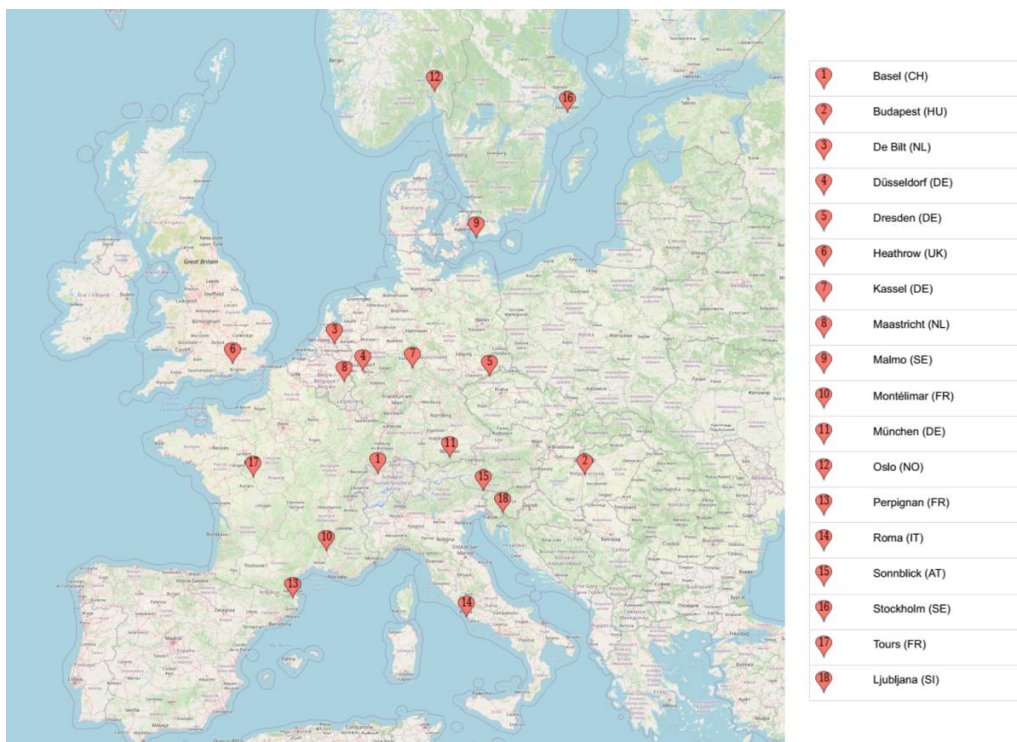


Figure 3. Locations of the 18 stations in the dataset (map provided by the dataset authors).

3. Methodology

3.1 Feature engineering

The feature vector contains 239 features. Current-day observations from all stations contribute 163 features. Two calendar features encode the day of year with a sine and cosine pair, which represents seasonality without treating December and January as distant values. Twenty-one features describe the recent history of Basel mean, minimum, and maximum temperature using lags of 1, 2, 3, and 7 days and rolling means over 7, 14, and 30 days. Fifty-one features capture lagged mean temperature from the other 17 stations, which allows the model to use spatial weather patterns. Two additional rolling features describe local pressure and humidity over the previous seven days. Every engineered feature for day t uses only day t and earlier observations.

The raw dataset provides only DATE (YYYYMMDD) and MONTH (1-12). The sine and cosine calendar encoding is added during feature engineering to represent the annual cycle continuously; it is not part of the source database.

3.2 Chronological evaluation protocol

The data are divided into three consecutive periods. The split is applied to the forecast target date rather than the feature date. This detail matters because a feature row for 31 December 2006 predicts 1 January 2007; placing that row in training would leak a validation target into the fitting set. Table 3 reports the resulting periods. The test period is 2009 plus 1 January 2010, which is the final target date available in the dataset.

Table 3. Chronological data splits.

Split	Samples	First target date	Last target date
Train	2,527	2000-01-31	2006-12-31
Validation	731	2007-01-01	2008-12-31
Test	366	2009-01-01	2010-01-01

3.3 Baselines

Two baselines define the minimum performance that a useful model should exceed. Persistence predicts the next day using the current day temperature, so y_{t+1} is approximated by y_t . This baseline is difficult to beat for temperature because daily temperature is highly auto-correlated. Monthly climatology predicts the average target value for the target calendar month, estimated on the fitting period. It represents a seasonal forecast that ignores the current weather state.

3.4 Models

In the model part of our project, we designed 4 models and finally kept the one with the best performance. With their experimental results evaluated respectively, four model families are compared. Ridge regression is a regularized linear model and provides an interpretable, low-variance reference. Random forest is a bagged ensemble of decision trees that can represent nonlinear relationships and

interactions. Histogram-based gradient boosting builds trees sequentially, with each tree correcting the remaining error of the current ensemble. A multi-layer perceptron provides a flexible neural baseline.

Ridge and the neural network use standardized inputs; tree-based models are scale invariant. Table 4 lists the tuning space and the configuration selected on the validation period for each family.

Table 4. Hyper-parameter search space and validation-selected configuration for each model family.

Model family	Validation search space	Selected configuration	Validation MAE (C)
Gradient boosting	learning_rate in {0.03, 0.05, 0.1}; max_leaf_nodes in {15, 31, 63}	learning_rate=0.05, max_leaf_nodes=31.0	1.217
Ridge regression	alpha in {0.1, 1, 10, 100}	alpha=10.0	1.287
Random forest	max_depth in {none, 10, 16}; min_samples_leaf in {1, 2, 4}	max_depth=16.0, min_samples_leaf=4.0	1.319
Neural network (MLP)	alpha in {1e-5, 1e-4, 1e-3}; hidden layers in {(64,), (128, 64)}	alpha=1e-05, hidden_layer_sizes=(128, 64)	1.365

3.5 Model selection and final evaluation

The training and tuning period of the model families made a through and detailed utilization of the data. Initially, each model family is tuned on the validation period and the family with the lowest validation MAE is selected as the final model. That model is then refitted on training plus validation data and evaluated once on the test period. Reporting every model on the test period keeps the comparison transparent, while the selection decision itself uses validation performance only. This design of flow ensures the models to be actually unfamiliar with the data in the testing data set, so that it avoids choosing a model because it happens to fit the test year.

3.6 Additional robustness check

A single test period can be sensitive to the weather of one particular year. An expanding-window backtest is therefore added. For each month of 2009, the selected model family is retrained on all data available before that month and used to predict the month. This produces twelve monthly evaluation windows and a second estimate of out-of-sample performance. The final one-day fold in January 2010 is reported for completeness but excluded from the monthly figure because it contains only one sample.

3.7 Metrics

MAE is the mean absolute difference between observed and predicted temperature. RMSE is the square root of the mean squared difference. R2 measures the fraction of variance explained relative to the mean of the test target. The skill score compares a model with persistence: $\text{skill} = 1 - \text{MAE}_{\text{model}} / \text{MAE}_{\text{persistence}}$. A positive skill score means the model improves on simply repeating the current temperature.

4. Experimental Setup

The pipeline uses Python with pandas (McKinney, 2010), NumPy (Harris et al., 2020), scikit-learn (Pedregosa et al., 2011), and matplotlib (Hunter, 2007). The random seed is 42 and all experiments run through a single entry point, `run_experiment.py`. The processed design matrix contains 239 features and 3,624 usable samples after removing the initial rows required to construct 30-day rolling features and the final row without a next-day target. The complete experiment, including tuning, final fitting, feature importance, and the expanding-window backtest, takes a few minutes running on my laptop.

Table 5. Experimental configuration.

Setting	Value
Target	BASEL_temp_mean on day t+1
Forecast horizon	1 day
Training period	Target dates up to 2006-12-31
Validation period	2007-01-01 to 2008-12-31
Test period	2009-01-01 to 2010-01-01
Number of features	239
Number of supervised samples	3,624
Primary metric	Mean absolute error (C)
Random seed	42

5. Results and Discussion

5.1 Test-period performance

Table 6. Test-period metrics for all baselines and models. Models are sorted by MAE.

Model	MAE (C)	RMSE (C)	R2	Skill vs persistence
Gradient boosting	1.228	1.619	0.957	0.263
Ridge regression	1.259	1.629	0.956	0.245
Random forest	1.292	1.658	0.954	0.225
Neural network (MLP)	1.306	1.672	0.954	0.217
Persistence	1.667	2.188	0.921	0.000
Monthly climatology	2.709	3.410	0.807	-0.625

The validation-selected model is Gradient boosting. On the held-out test period it achieves an MAE of 1.228 C, an RMSE of 1.619 C, and an R2 of 0.957. The persistence baseline has an MAE of 1.667 C, so the selected model reduces the average absolute error by 26.3 percent.

Overall, all four learned models beat the persistence baseline, while monthly climatology performs substantially worse because it cannot react to the current weather state. The cleaned-data ranking is stable: gradient boosting is followed by ridge regression, random forest, and the neural network. The neural-network result is worth highlighting because it exposes the value of the data audit. Before cleaning, the selected MLP configuration had a validation MAE of 1.826 C, an RMSE of 7.466 C, an R2 of -0.192, and a maximum error of 163.3 C, with three errors above 50 C. The largest error was traced to TOURS_pressure = 0.0003 on 2008-12-30, which is 121.5 standard deviations below the training mean. After physical-range cleaning, the same model family reaches a validation MAE of 1.365 C, an RMSE of 1.748 C, an R2 of 0.935, and a maximum error of 7.56 C, with no errors above 50 C. Tree models were insensitive to the invalid input because they cannot extrapolate beyond their leaf ranges, while the neural network could. This is a useful robustness lesson and is why the cleaned validation table is the one used for model selection.

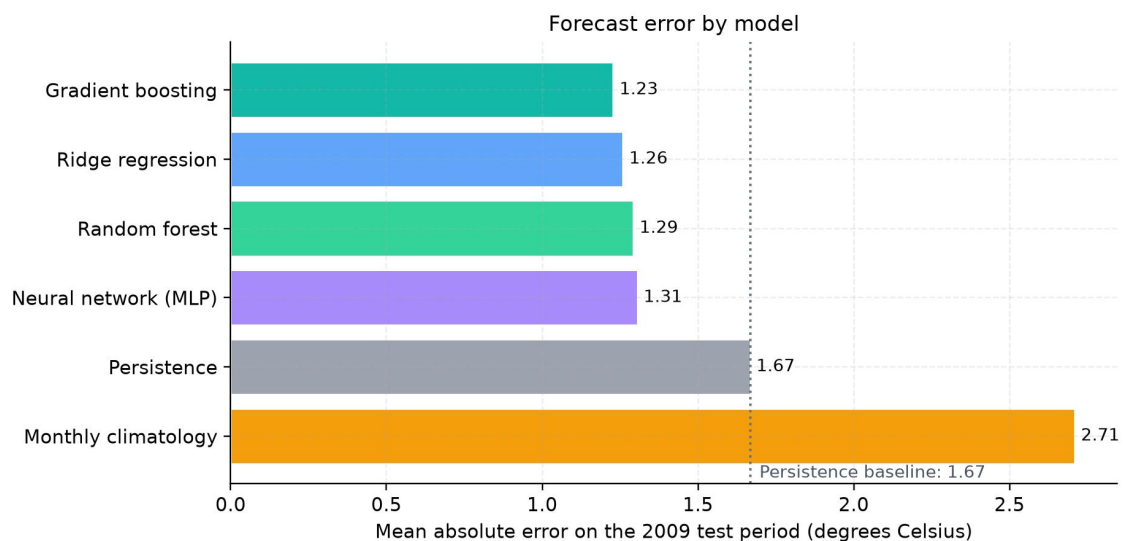


Figure 4. Mean absolute error on the 2009 test period. The dotted line is the persistence baseline.

5.2 Forecast behaviour

The selected model tracks the seasonal cycle and the short-term variability of the test period. Figure 5 shows seven-day smoothed forecasts so that the seasonal signal and the main warm and cold episodes remain visible. Figure 6 shows the relationship between predicted and observed values. The points cluster around the diagonal, which indicates that the model captures the overall level and the day-to-day variation. The largest remaining errors occur during rapid transitions, such as the spring warming and the onset of winter, when the next day can depart quickly from the current state.

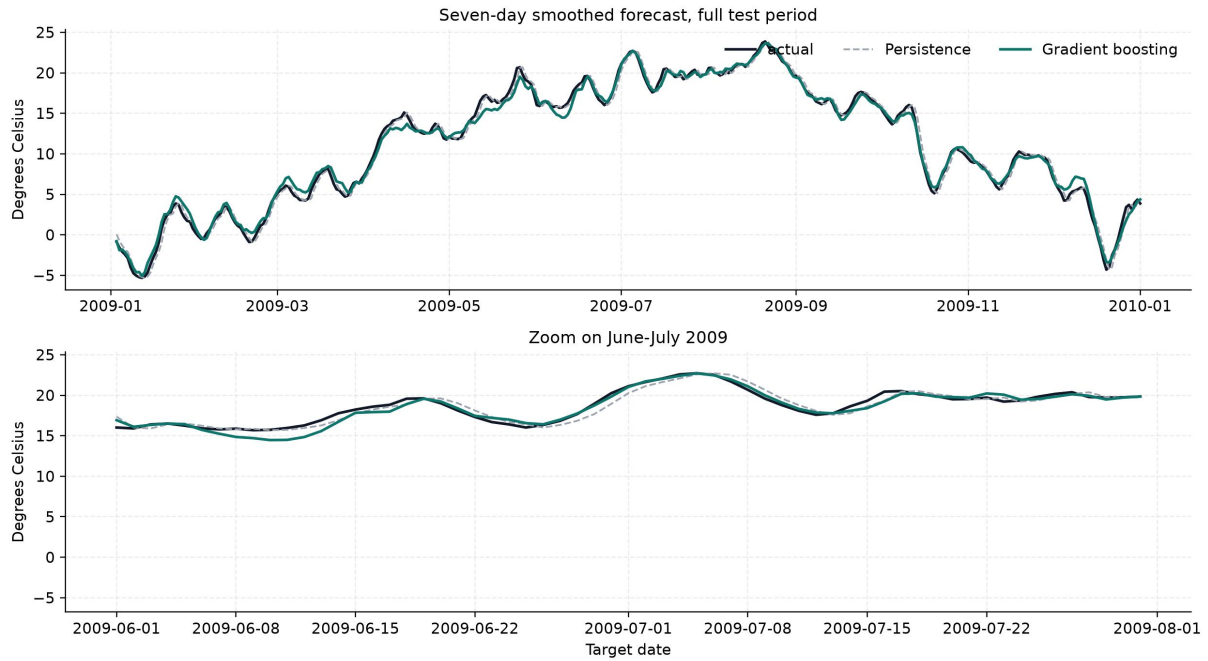


Figure 5. Seven-day smoothed observed and predicted Basel mean temperature over the test period.

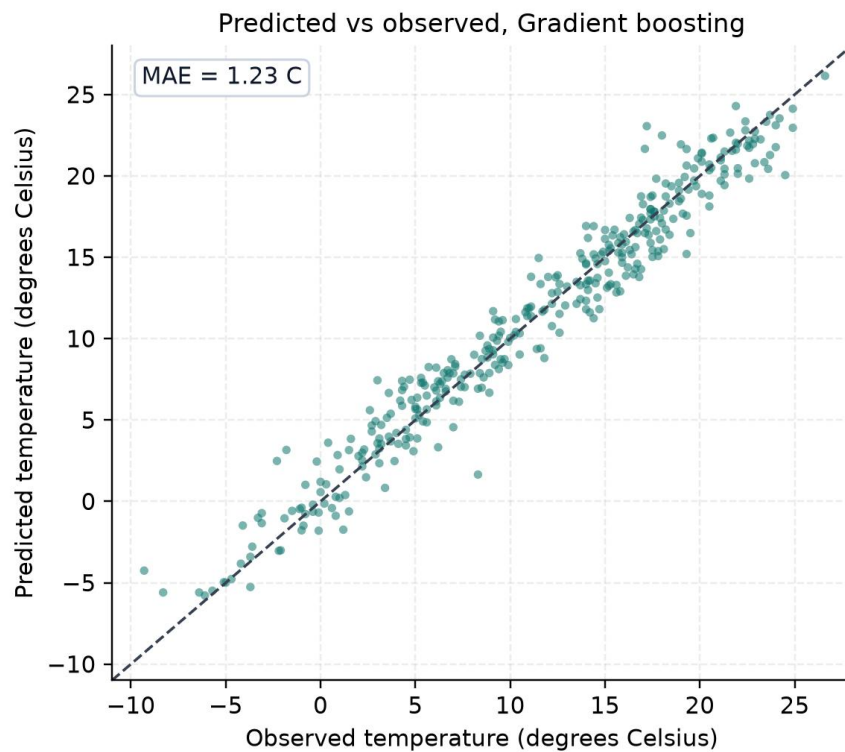


Figure 6. Predicted versus observed temperature for the selected model on the test period.

5.3 Seasonal error analysis

Figure 7 and Table 7 break the test error down by calendar month. The model improves on persistence in every month of the 2009 test period. Its largest absolute errors occur in May and December, while September and October are the easiest months. This pattern is consistent with the physical behaviour of temperature: stable autumn conditions are more stable and rarely fluctuates, so that to be much easier to predict, whereas spring and early winter contain larger and more variable day-to-day changes,

resulting in more significant prediction errors. The result also shows why a single overall score is not enough; the value of the model depends on the season, and the rolling backtest in Section 5.5 provides a second view in which months like April are essentially tied with persistence.

Table 7. Monthly mean absolute error on the test period. Improvement is relative to persistence.

Month	Model MAE (C)	Persistence MAE (C)	Improvement
1	1.204	1.541	21.9 percent
2	1.064	1.768	39.8 percent
3	1.252	1.406	11.0 percent
4	1.468	1.570	6.5 percent
5	1.758	1.881	6.5 percent
6	1.250	1.453	14.0 percent
7	1.132	1.829	38.1 percent
8	1.062	1.948	45.5 percent
9	0.770	1.113	30.8 percent
10	0.969	1.632	40.6 percent
11	1.100	1.423	22.7 percent
12	1.681	2.419	30.5 percent

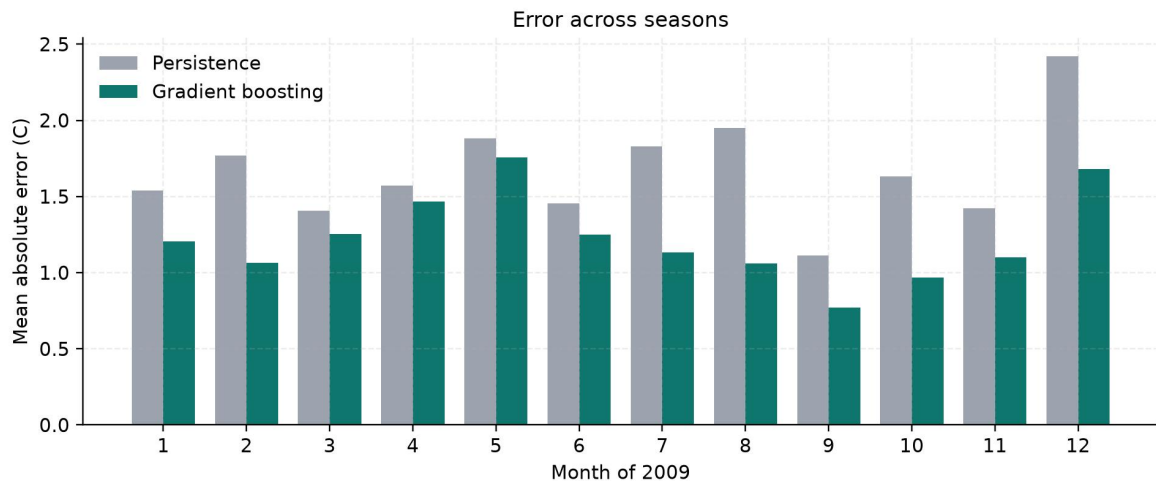


Figure 7. Monthly mean absolute error for the selected model and the persistence baseline.

5.4 Feature importance

Permutation importance measures how much the test error increases when a feature is randomly shuffled. The current-day Basel mean temperature is by far the most important feature, which is expected because it is the persistence signal. Spatial information adds measurable value: mean temperature at Maastricht and maximum temperature at Tours and Heathrow appear near the top of

the ranking. Calendar seasonality and local minimum temperature also contribute. This pattern supports the modelling choice of combining local persistence with cross-station and seasonal context rather than relying on a single station or a single variable.

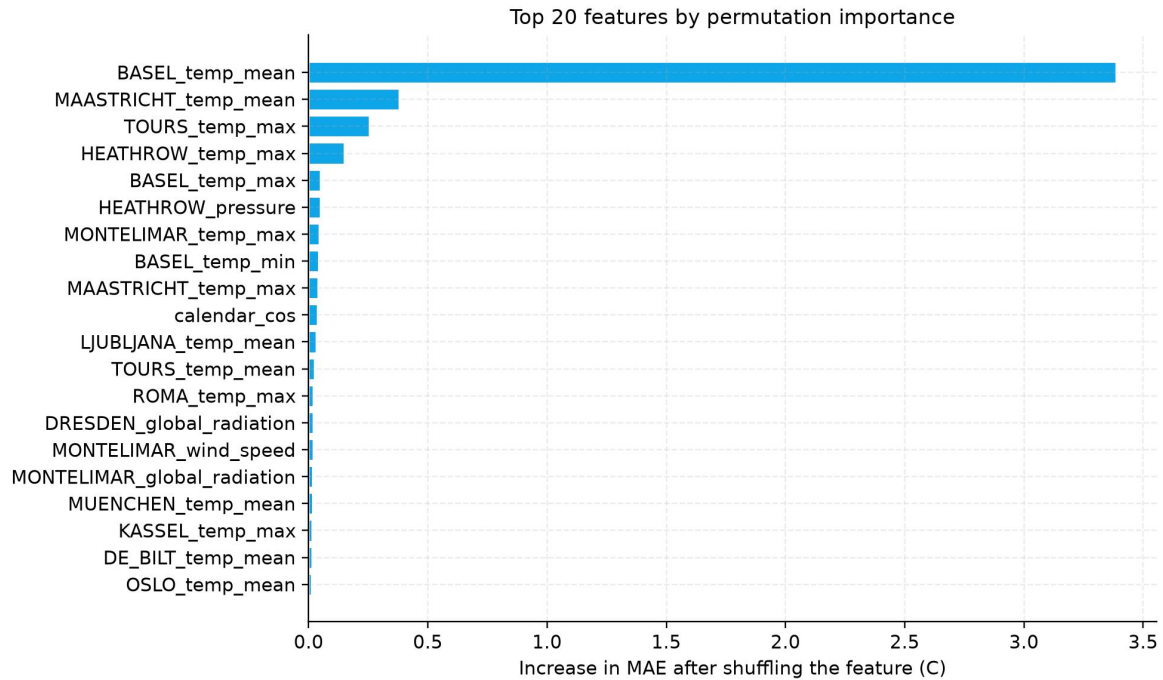


Figure 8. Top 20 features by permutation importance for the selected model.

5.5 Expanding-window robustness check

The expanding-window backtest retrains the selected family before each month of 2009 and therefore uses less data for the early months. Across the full backtest of 366 predictions, including the final one-day fold, the model achieves an MAE of 1.219 C compared with 1.667 C for persistence, a skill score of 0.269. Figure 9 shows the monthly results. The conclusion is consistent with the fixed test-period evaluation: the model improves on persistence in most months, while April is essentially tied with the baseline and the largest absolute errors remain concentrated in spring and early winter.

Table 8. Monthly expanding-window backtest results.

Fold	Samples	Model MAE (C)	Persistence MAE (C)	Skill
2009-01	31	1.177	1.471	0.200
2009-02	28	1.067	1.768	0.397
2009-03	31	1.221	1.406	0.132
2009-04	30	1.593	1.570	-0.015
2009-05	31	1.575	1.881	0.162
2009-06	30	1.216	1.453	0.163
2009-07	31	1.102	1.829	0.397
2009-08	31	1.104	1.948	0.433
2009-09	30	0.751	1.113	0.325
2009-10	31	0.944	1.632	0.422
2009-11	30	1.151	1.423	0.191
2009-12	31	1.664	2.419	0.312

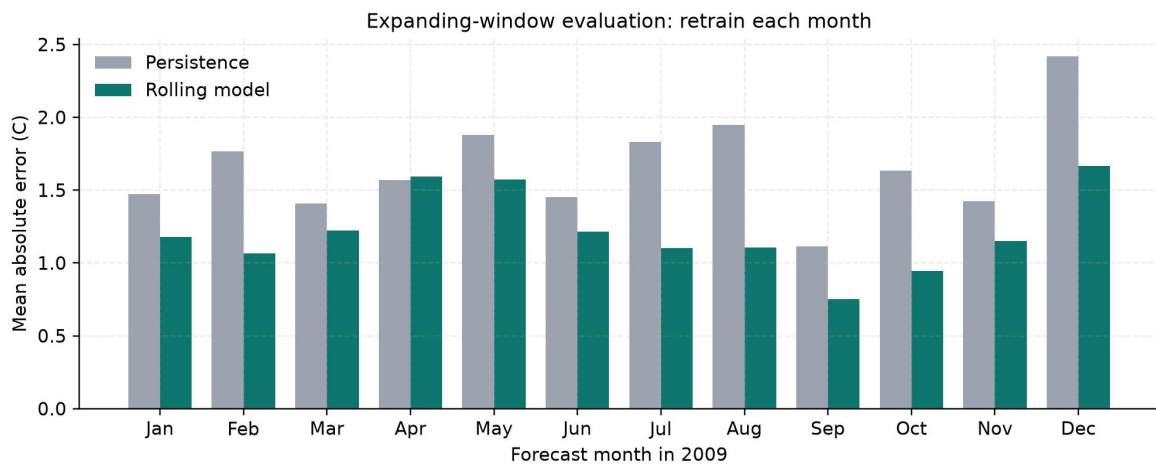


Figure 9. Expanding-window backtest: the model is retrained before each month of 2009.

6. Limitations and Future Work

The project demonstrates a complete and defensible forecasting workflow, but several limitations should be stated. The model predicts a single station and a single lead time. It produces a point forecast rather than a calibrated uncertainty interval. The dataset is daily and station-based, so it does not represent the full spatial structure of the atmosphere. The released data contain no NaN or -9999 values, but the physical-range audit found 36 impossible values in the pressure, cloud-cover, and sunshine series. The pipeline cleans these values explicitly, but the need for such a check shows that the released data are not entirely free of quality problems, and operational systems would also need to handle delayed reporting. The record covers only about ten years, and the test period is one specific year. Finally, the reported

metrics assume that all observations for day t are available at prediction time, while in actual practice, a real deployment would need to respect station reporting delays.

Several extensions would strengthen the work if more specific requirements are needed. A probabilistic model, such as quantile regression or a distributional neural network, would produce uncertainty intervals that are useful for decision making. Multi-horizon forecasting would extend the setup from one day to several days ahead. Multi-station and multi-target models could share information more efficiently than a separate model per station. Spatiotemporal models, including graph neural networks or sequence models, could represent the spatial dependence between stations more directly. External data such as reanalysis fields, satellite observations, or elevation could improve performance. Finally, due to potential scenarios about commercial and publication cases, a proper operational evaluation would include delayed reporting, missing values, and distribution shift over time.

7. Reproducibility

The project is organized so that the reported results can be regenerated from the included data and code. The repository contains a data directory, a source package, a single experiment entry point, and directories for metrics, predictions, and figures. The experiment uses a fixed random seed and writes all reported numbers and figures to the outputs directory. Here is a project overview structure of our work:

```
weather_prediction_project/
  data/                # original dataset files
  src/                 # data, features, models, plots, experiment
  outputs/metrics/     # metrics and summaries
  outputs/predictions/ # test and rolling predictions
  outputs/figures/     # report figures
  outputs/models/      # saved model and metadata
  run_experiment.py    # main entry point
```

To reproduce the experiment,

install the dependencies listed in requirements.txt and run `python run_experiment.py`

The script rebuilds the feature matrix, tunes each model family, selects the final model on validation performance, evaluates it on the test period, runs the expanding-window backtest, and writes all figures, metrics, predictions, and the fitted model to the outputs directory. All data contents of result are easily to access for upcoming application and utilization.

8. References

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Appendix A. Validation Metrics

Table A1. Validation-period metrics. Model selection uses this table; the test period is not used for selection.

Model	MAE (C)	RMSE (C)	R2	Skill vs persistence
Gradient boosting	1.217	1.546	0.949	0.258
Ridge regression	1.287	1.612	0.944	0.215
Random forest	1.319	1.661	0.941	0.195
Neural network (MLP)	1.365	1.748	0.935	0.167
Persistence	1.639	2.085	0.907	0.000
Monthly climatology	3.164	3.876	0.679	-0.930

Appendix B. Top Feature Importances

Table B1. Top 20 features by permutation importance on the test period.

Rank	Feature	Importance (C)	Standard deviation
1	BASEL_temp_mean	3.386	0.150
2	MAASTRICHT_temp_mean	0.381	0.008
3	TOURS_temp_max	0.255	0.008
4	HEATHROW_temp_max	0.150	0.012
5	BASEL_temp_max	0.048	0.005
6	HEATHROW_pressure	0.048	0.012
7	MONTELMAR_temp_max	0.044	0.005
8	BASEL_temp_min	0.041	0.006
9	MAASTRICHT_temp_max	0.039	0.010
10	calendar_cos	0.036	0.010
11	LJUBLJANA_temp_mean	0.031	0.003
12	TOURS_temp_mean	0.025	0.001
13	ROMA_temp_max	0.020	0.001
14	DRESDEN_global_radiation	0.019	0.005
15	MONTELMAR_wind_speed	0.018	0.005
16	MONTELMAR_global_radiation	0.018	0.005
17	MUENCHEN_temp_mean	0.017	0.003
18	KASSEL_temp_max	0.014	0.002
19	DE_BILT_temp_mean	0.014	0.002
20	OSLO_temp_mean	0.012	0.002